



# Testing models for the formation of the equatorial ridge on Iapetus via crater counting



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## ABSTRACT

Iapetus's equatorial ridge, visible in global views of the moon, is unique in the Solar System. The formation of this feature is likely attributed to a key event in the evolution of Iapetus, and various models have been proposed as the source of the ridge. By surveying imagery from the Cassini and Voyager missions, this study aims to compile a database of the impact crater population on and around Iapetus's equatorial ridge, assess the relative age of the ridge from differences in cratering between on ridge and off ridge, and test the various models of ridge formation. This work presents a database that contains 7748 craters ranging from 0.83 km to 591 km in diameter. The database includes the study area in which the crater is located, the latitude and longitude of the crater, the major and minor axis lengths, and the azimuthal angle of orientation of the major axis. Analysis of crater orientation over the entire study area reveals that there is no preference for long-axis orientation, particularly in the area with the highest resolution. Comparison of the crater size-frequency distributions show that the crater distribution on the ridge appears to be depleted in craters larger than 16 km with an abruptly enhanced crater population less than 16 km in diameter up to saturation. One possible interpretation is that the ridge is a relatively younger surface with an enhanced small impactor population. Finally, the compiled results are used to examine each ridge formation hypothesis. Based on these results, a model of ridge formation via a tidally disrupted sub-satellite appears most consistent with our interpretation of a younger ridge with an enhanced small impactor population.

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## 1. Introduction

Iapetus, Saturn's third largest moon, is the only known body in the Solar System with a mountainous equatorial ridge. The ridge extends >75% of the circumference of the satellite (though not continuously) and has been heavily modified by impacts and landslides. Detailed topographic analysis confirmed that the peaks on the ridge are variable, ranging from sharp steep peaks, flat tops (trapezoidal), or numerous parallel ridge crests (Singer et al., 2009; Singer and McKinnon, 2011; Singer et al., 2012; Lopez Garcia et al., 2014). The formation of the ridge is likely attributed to a key event in the evolution of Iapetus. Studying the ridge can yield information about the formation, evolution, and the history of Iapetus. Various models (reviewed in Section 2.1) have been proposed as the source of the ridge, all of which fall under two broad categories: endogenic or exogenic. In this study, we test these formational models by exploring the population of impact craters on the sur-

face of Iapetus as each ridge formation model makes predictions about the expectant crater population.

Studying impact craters and their distribution is a primary tool used to learn about the geologic history of planetary bodies. If the crater population is well constrained, it can then be used to determine relative ages, absolute model ages (if the impacting projectile population is known), resurfacing rates, to provide clues about subsurface properties, and to explore the evolution of the surface and planetary body as a whole (e.g., Melosh, 1989). To date, only a few groups have published their work on the impact craters and their distribution on Iapetus (Denk et al., 2010; Hirata, 2016; Kirchoff and Schenk, 2009, 2010; Martin and Jurdy, 2010; Neukum et al., 2005; Rivera-Valentin et al., 2014; Schmedemann et al., 2008, 2009). In this study, we present a crater database that includes 7748 craters on and around Iapetus's ridge. We then use the database to investigate the qualitative and quantitative characteristics of the crater population to reconstruct the age of the ridge relative to the surrounding terrain. Finally, we delve deeper into the compiled results to examine each ridge formation hypothesis and discuss the likelihood of the ridge formation models.

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#### 4. Discussion

Table 3 shows the summarized results for this study. The original hypothesis matrix (Table 1) includes Modified Craters as a possible identifier in the crater population. Visual inspection of the craters revealed no evidence of tectonic, volcanic, or surface resetting beyond crater saturation or landslides. Discussed below are the implications for each ridge formation hypotheses in combination with the summarized results.

In regions with the most fidelity (i.e., the High Resolution Region), the size frequency distributions show that on the ridge, there is a decrease in larger craters ( $D > 16$  km) with a rather abrupt enhancement in smaller sizes ( $D < 16$  km). In general, the ridge may be systematically depleted in larger craters for most terrains, though the uncertainties and possible saturation render this conclusion equivocal. If this observation is real and not due to other factors (e.g., observational biases), this abrupt change at  $\sim 16$  km is difficult to reconcile with continuous crater populations. Observed crater populations of any common size-frequency distribution slope typically fall away from saturation at smaller craters gradually, not abruptly (Hartmann, 1984; Chapman and McKinnon, 1986). One possible interpretation of the data is that the ridge is a relatively younger surface with an enhanced small impactor population from a non-heliocentric source, thus allowing large craters to sit below saturation but with an abrupt transition to saturation at small sizes. The only current model that produces conditions with an enhanced small, non-heliocentric impactor population and depleted in larger craters preferentially along the ridge is the exogenic, impact generated sub-satellite model by Dombard et al. (2012). Here, the ridge can be much younger, explaining the lower degree of cratering at mid to large sizes, while the enhanced impactor flux arising from very comminuted, collisionally evolved debris from a tidally disrupted sub-satellite would explain the abrupt enhancement of small craters on the ridge.

Placing a younger age on the ridge would rule out all current endogenic ridge formation models, as the endogenic ridge formation models place the age of the ridge early on in Iapetus's history. In addition, most endogenic models would result in some form of crater transformation, which we defined by any crater subsequently overprinted by geologic processes. No evidence of modified craters is found in the crater population. Even if geologic processes, such as tectonics and volcanism did occur, there would be preferential removal of smaller craters, not larger ones.

With the exception the Dombard et al. (2012) model, the current exogenic models, primordial ring system (Ip, 2006) and impact generated ring system (Levison et al., 2011; Kuchta et al., 2015), also place the age of the ridge early on in Iapetus's history. An ancient ring system should produce a ridge that is only slightly (by a matter of centuries or millennia) younger in age; however, the expected enhanced impactor population may make the ridge more cratered and thus appear older. Both exogenic models that involve a giant impact should produce a ridge that is younger in age, although the ridge in the Levinson model would, like in the Ip model, only be centuries to millennia younger while in the Dombard model, it could conceivably being hundreds of millions to a billion years younger, depending on the despinning time scale of Iapetus.

Furthermore, the low speed, shallow impacts from the exogenic models might produce ragged craters preferentially elongated in the E-W direction along the equator. The rose diagrams did not indicate that the surface of Iapetus displays a preference for craters oriented in the E-W direction. However, one noticeable feature in the rose diagrams concerns craters on the ridge, which tend to be oriented obliquely between N-S and E-W. This concentration at oblique angles, however, is likely due to observational bias. The ridge is a three-dimensional structure viewed obliquely, the im-

ages from which are map projected in the basemaps. As a consequence, angles are affected. As evidence, the landslides shown in Fig. 3 should run N-S, but do not in the basemaps. Thus, the orientation of the long axes of the craters likely does not display a robust signal associated with the ridge and the orientation data cannot be used to discern any particular model.

The issue of crater saturation has been discussed in the literature extensively (e.g., Gault, 1970; Hartmann, 1984; Melosh, 1989; Richardson, 2009). The spatial density of impact craters reaches saturation equilibrium conditions with older terrains first and is reflected in crater counts that level off as a function of time and further bombardment. If crater saturation has not been reached for a particular crater diameter, then counting larger craters can form the basis for an estimate of the age of the surface. Fig. 4 indicates that all terrain units on Iapetus likely reach or hover just below saturation at  $D > 10$  km, within error. In regions with the most fidelity (i.e., High Resolution Region and Leading Hemisphere), the R-plots show a slight decrease in larger sized craters with an increase in smaller sizes up to saturation on the ridge, while the rest of the off-ridge surface is likely sitting at saturation. This issue of saturation coupled with the relatively large errors on the counts, however, limits the robustness of any conclusions drawn from this study.

#### 5. Conclusions

Iapetus is the only known body in the Solar System that has a mountainous equatorial ridge, and the formation of this feature is likely attributed to a key event in the evolution of the satellite. Various models have been proposed as the source of the ridge, all of which include predictions about the expected crater population. The objectives of this study have been to compile a database of the crater population on and around the equatorial ridge, to assess the relative age from differences in cratering between on ridge and off ridge, and to test the various models of ridge formation.

The size-frequency distributions of the craters are used to estimate relative terrain ages, interpret impactor population source, identify resurfacing events, and assess whether saturation has been reached. In regions with the most fidelity, the results may show a depletion of larger craters on the ridge with an enhancement in smaller sizes up to saturation. While the ridge does appear less cratered at larger sizes, the potential for surface resetting (i.e., via landslides) within close proximity to the ridge and the combination of large uncertainties and crater counts near saturation render these conclusions equivocal. Analysis of crater orientation reveal that overall, there is no preference for crater major-axis orientation.

Based on the results from the statistical analysis, it is concluded that the exogenic impact generated ridge formation model proposed by Dombard et al. (2012) is the most consistent with the observed crater population in this study, as it is the only current model that produces conditions with an enhanced small, non-heliocentric impactor population, yet can result in the ridge under populated in larger craters relative to off ridge (and hence be younger). While this study provides a broad overview of the craters on Iapetus, higher resolution imagery and further analysis of the crater population, such as morphologic classification, are needed to state definitively which ridge formation mechanism is most viable.

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